MIL-S-19500/122C
31 JANUARY 1967
SUPERSEDING
MIL-S-19500/122B
28 July 1965

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, HIGH-POWER

TYPE 2N1358

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a high-power, PNP, germanium transistor.

1.2 Physical dimensions. See figure 1 (TO-36).

1.3 Maximum ratings.

P_C 1/ T_{MB} = 25°C	I_E	I_B	V_{CBO}	V_{EBO}	V_{CEO}	T_{stg}
W	Adc	Adc	Vdc	Vdc	Vdc	°C
150	15	4	-80	-40	-40	-65 to +100

1/ Derate 2.0 W/°C for T_{MB} > 25°C.

1.4 Primary electrical characteristics.

	h_{FE} at V_{CE} = -2 Vdc		V_{CE}(sat) I_C = -12 Adc I_B = -2 Adc	f_{hfe}	θ_{J-C}
	I_C = -1.2 Adc	I_C = -5 Adc	Vdc	kHz	°C/W
Min	40	25	---	5	---
Max	100	50	-0.7	---	0.5

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

FSC 5961

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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

T_{MB} . . . . . Temperature of the mounting surface of the device.

3.3 Design, construction, and physical dimensions. Transistor shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

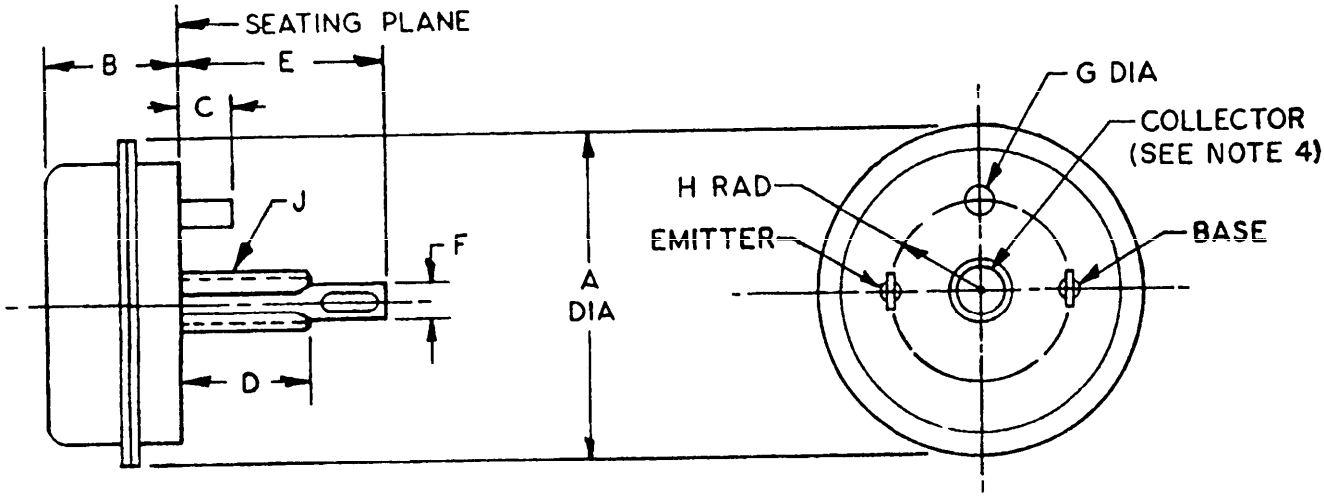
4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	---	1.250	---	31.75	
B	---	.520	---	13.21	
C	.125	.312	3.18	7.92	
D	.375	.500	9.53	12.70	
E	.610	.710	15.49	18.03	3
F	---	.190	---	4.83	3
G	---	.140	---	3.56	5
H	.335	.355	8.51	9.02	
J	---	---	---	---	2

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. 10-32 UNF-2A, according to Handbook H28.
3. Two leads.
4. The collector shall be internally connected to the mounting base.
5. Cylindrical surface of the locating pin shall be insulated so that electrical contact is not made with the heat sink. Dimension G shall include this insulation. Figure 2 preferred measurement method.

FIGURE 1. Physical dimensions of transistor type 2N1358 (TO-36).

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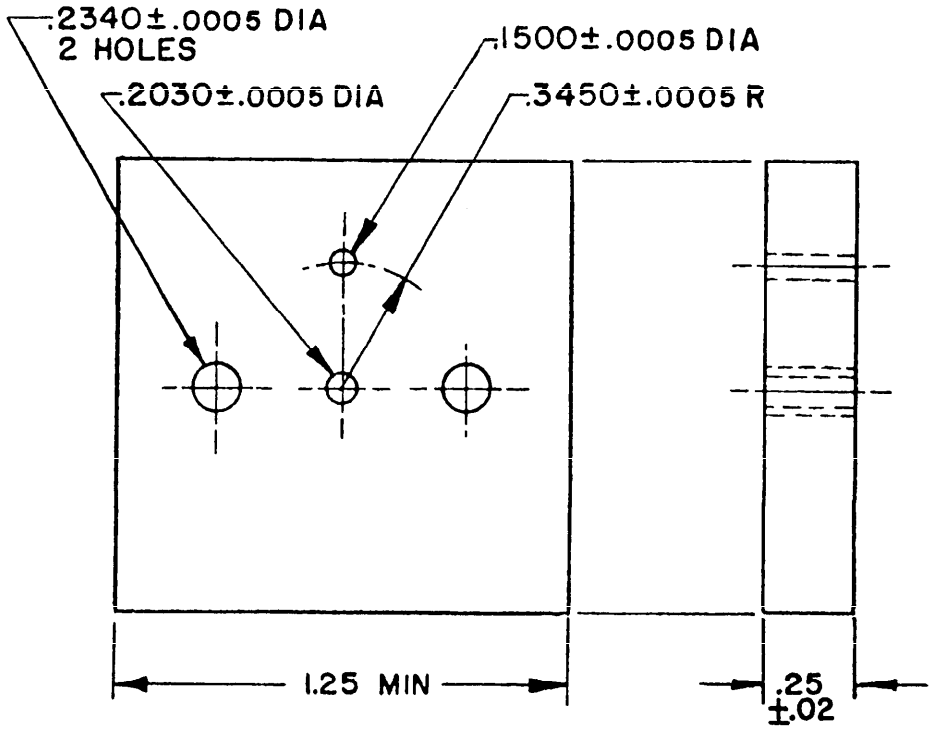


FIGURE 2. Alinement gage for transistor type 2N1358.

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* 4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hours life-test, may be continued on test for 1, 000-hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1, 000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1, 000 hours shall be computed for 1, 000-hour acceptance criteria.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

4.4.1 Inspection conditions. All measurement shall be made at T_{MB} = 25°C unless otherwise specified.

* 4.4.2 Terminal strength (stud torque). Acceptance criteria after the stud torque test shall be 10-32 UNF-2A for external threaded parts in accordance with Handbook H28.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Breakdown voltage, collector to emitter	3011	Bias cond. D; I_C = -300 mAdc		BV_{CEO}	-40	---	Vdc
Breakdown voltage, collector to emitter	3011	Bias cond. C; I_C = -300 mAdc		BV_{CES}	-70	---	Vdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -2 Vdc		I_{CBO}	---	-200	µAdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -80 Vdc		I_{CBO}	---	-4	mAdc
Emitter to base cutoff current	3061	Bias cond. D; V_{EB} = -40 Vdc		I_{EBO}	---	-4	mAdc
* Subgroup 3			5				
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -1.2 Adc; t_p = 1 sec max		h_{FE}	40	100	---
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -5 Adc; t_p = 1 sec; max, duty cycle = 1 to 2%		h_{FE}	25	50	---
Collector to emitter voltage (saturated)	3071	I_C = -12 Adc; I_B = -2 Adc		$V_{CE}(sat)$	---	-0.7	Vdc
Base emitter voltage (nonsaturated)	3066	Test cond. B; V_{CE} = -2 Vdc; I_C = -1.2 Adc		V_{EB}	---	0.5	Vdc
Base emitter voltage (nonsaturated)	3066	Test cond. B; V_{CE} = -2 Vdc; I_C = -5 Adc		V_{EB}	---	0.9	Vdc

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TABLE I. Group A inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
* Subgroup 4			10				
Small-signal short-circuit forward-current transfer-ratio cutoff frequency	3301	V_{CE} = -6 Vdc; I_C = -5 Adc		f_{hfe}	5	---	kHz
High-temperature operation	---	T_C = +71°C min		---	---	---	---
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-6	mAdc

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			15				
Solderability	2026	Omit aging		---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. B, except step 3; t_{max} = +95 $^{+5}_{-0}$ °C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. B		---	---	---	---
Terminal strength (tension)	2036	Test cond. A; weight = 10 lbs; t = 15 sec to each terminal		---	---	---	---
Terminal strength (terminal torque)	2036	Test cond. D1; torque = 24 in.-oz. to be applied to flat of each terminal for t = 15 sec		---	---	---	---
Terminal strength (stud torque)	2036	Test cond. D2; torque = 12 in.-lbs.; t = 15 sec (see 4.4.2)		---	---	---	---
Seal (leak-rate)	---	Method 112 of MIL-STD-202, test cond. C, procedure III; test cond. B for gross leaks		---	---	$5X10^{-7}$	atm cc/sec
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -80 Vdc		I_{CBO}	---	-4	mAdc
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -5 Adc; t_p = 1 sec max; duty cycle = 1 to 2%		h_{FE}	25	50	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 3			15				
Shock	2016	Nonoperating; 1500 G, approx. 0.5 msec; 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration fatigue	2046	Nonoperating		---	---	---	---
Vibration, variable frequency	2056			---	---	---	---
Constant acceleration	2006	Nonoperating; 2000 G in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			20				
* Salt atmosphere (corrosion)	1041			---	---	---	---
End points: (Same as subgroup 2)							
* Subgroup 5			7				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C; time = 340 hours		---	---	---	---
End points: Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -80 Vdc		I_{CBO}	---	-8	mAdc
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -5 Adc; t_p = 1 sec max, duty cycle = 1 to 2%		h_{FE}	20	60	---
* Subgroup 6			10				
Steady state operation life	1026	V_{CB} = -15 Vdc; P_C = 20 W; T_{MB} = +90°C; time = 340 hours		---	---	---	---
End points: (Same as subgroup 5)							

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			20				
Barometric pressure, reduced (altitude operation)	1001	Normal mounting; Pressure = 8 mm. Hg. for 60 sec min.		---	---	---	---

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TABLE III. Group C inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1 - Continued							
Measurement during test:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -80 Vdc		I_{CBO}	---	-4	mAdc
Thermal resistance	3151			θ_{J-C}	---	0.5	°C/W
* Floating potential	3020	V_{CB} = -80 Vdc; voltmeter input resistance ≥ 10 meg		V_{EBF}	---	-0.5	Vdc
Subgroup 2			20				
Burnout by pulsing	3005	I_B = -4 Adc; I_E = 15 Adc; t_p = 60 sec min 1 cycle		---	---	---	---
End points: Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -5 Adc; t_p = 1 sec max, duty cycle = 1 to 2%		h_{FE}	25	50	---
Subgroup 3			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C		---	---	---	---
End points: (Same as subgroup 5 of group B)							
Subgroup 4			λ = 15				
Steady state operation life	1026	V_{CB} = -15 Vdc; P_C = 20 W; T_{MB} = +90°C		---	---	---	---
End points: (Same as subgroup 5 of group B)							

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Changes from previous issue. The outside margins of this document have been marked "*" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 11

Review activities:
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User activities:
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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

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1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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